A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 28

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors – many rare species with small ranges **1** High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) **1** Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches – intense competition for light drives specialisation **1** Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships – plants flower and fruit at different times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Adapted to stable conditions so they have narrow ranges of tolerance – a small change can move conditions outside the range **1** Many species are rare with small ranges / small populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) **1** Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Over half the Amazon's rainfall comes from transpiration within the forest – clearance reduces downwind rainfall **1** Rapid decay recycles nutrients so soils are shallow / poor – canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1 Ecological footprint:** the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced **1 Biocapacity:** the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** A hectare of land / sea with the world-average biological productivity **1** Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 8.1 / 1.7 = 4.8 (accept 4.7-4.8) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Earths) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint **1** Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO2 accumulates (overshoot) **1** Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up – it falls earlier each year (8 August in 2016) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Renewable energy and energy conservation (carbon footprint is the largest component) **1** Reduced meat consumption / food waste – local seasonal food **1 Circular economy:** repair, re-use, recycling – longer product lives **1** Public transport, efficient buildings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **energy conservation** → using less energy, eg by insulation or behaviour change • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable **1** Wild species have different ranges of tolerance so cress results cannot be applied directly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **tolerance** → the range of conditions in which a species can survive **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Preliminary study with widely spaced pH values to find where germination occurs **1** Final range from below the lowest pH giving germination to above the highest, with closer intervals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** Temperature (constant, natural germination temperature) **1** Light, water volume, number of seeds per dish, seed source **1** Growth medium (filter paper; soil would alter pH) – nutrients not needed for germination **1** Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface water affects fresh mass and drying damages the plants – dry mass requires killing the plants so the same plants cannot be re-measured **1** Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1
04.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1 Primary:** succession on a newly created habitat with no life / no soil (bare rock, new pond, sand) **1 Secondary:** succession restarting after an existing community is disturbed (tree felling, ploughing, fire, end of grazing) **1** Faster because soil already exists and many seeds / roots / organisms are still present so there are fewer delays for colonisation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Human activity (grazing, mowing, burning, coppicing) regularly removes the later successional species so the climatic climax cannot form **1** The stable community that develops under this repeated disturbance is a plagioclimax **1 Examples:** chalk grassland / hay meadows maintained by grazing or mowing – lowland heath by burning – coppiced woodland – Machair grassland – Broads grazing marshes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deflected succession** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** If the traditional activity stops, secondary succession returns the habitat to scrub / woodland and the plagioclimax species are lost **1** Rainforest and coral reefs are climax communities so conservation aims to minimise human impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Extreme environments (deserts, polar) are dominated by abiotic factors – few species tolerate them – populations fluctuate and food webs are simple so changes in one food species have large effects **1** Stable environments allow many species to evolve inter-species relationships – populations are regulated by biotic factors – complex food webs give stability (rainforests, reefs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 76-77; AQA spec 3.1.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Potato bean (N America): high protein content **1** Morama bean (southern Africa): protein content similar to soya, drought resistant **1** Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils **1** Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion **1 American bison:** graze grassland too poor for cattle **1 Common eland:** browse and graze so can be kept with cattle without severe competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil **1** So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	1 mark for the point below. **1** They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (15 - 6) / 6 x 100 **1** = 150% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cells shed by the newts into the water contain their DNA **1** Water samples are analysed for DNA sequences unique to the species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Catches depend on trapping effort, weather and newt activity on the night, so they vary from year to year **1** Trapping only samples one pond and one part of the population, whereas eDNA surveys all 20 ponds with a standard method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Image databases of unique markings (tiger stripes, whale shark spots, dolphin fins) giving territory size, movements, lifespan and social groups – limited to species with unique features • Motion-sensitive and infrared cameras, CCTV of nests with less disturbance – marking (rings, wing tags) • **DNA:** eDNA to detect presence (great crested newts, invasive carp) – DNA databases to identify regional populations and trace timber, ivory and fish • **Auditory monitoring:** bat detectors and sonograms for presence, abundance and activity • **Position monitoring:** radio, GPS and acoustic transmitters – geolocator tags for small birds – data recorders – carrier systems - ROVs / drones, AUVs, balloons, aircraft, satellites, animals • **Satellites:** visible light for land use and deforestation, infrared for vegetation and temperature, radar for wind, waves, sea level and oil – GPS for tracking • **Evaluation:** greater coverage, less disturbance and continuous data vs cost, battery / weight limits, calibration, need for ground-truthing, ethical issues of tagging – indirect evidence and traditional methods still needed • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	9
	Do not credit • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	New technologies have transformed what can be monitored and how often. Image databases of unique markings - tiger stripes, whale-shark spots, dolphin fin scars - let individuals be recognised, giving territory size, movements, lifespan and social structure without repeated capture, though only for species with unique features. Motion-sensitive and infrared cameras and CCTV record activity day and night with far less disturbance than observers, especially at nests.	
	Genetics adds detection where nothing can be seen: eDNA in water reveals great crested newts or invasive carp from shed cells, and DNA databases identify which regional population timber, ivory or fish came from, supporting enforcement. Bat detectors and sonograms give presence, abundance and activity of species heard but not seen. Position monitoring uses radio, GPS and acoustic transmitters, or tiny geolocator tags for birds too small to carry a transmitter, while data recorders on drones, AUVs, balloons, aircraft, satellites or the animals themselves collect abiotic data. Satellites map land use and deforestation in visible light, vegetation density and temperature in infrared, and wind, waves, sea level and oil slicks with radar that works through cloud.	
	The gains are coverage, continuity and reduced disturbance. The limits are cost, battery life and weight, the need to calibrate sensors and ground-truth remote data, ethical questions about tagging, and the fact that many species still require traditional methods and indirect evidence. New technology therefore complements rather than replaces fieldwork, but for rare, remote or nocturnal species it has made effective conservation possible where it was not before.	
	Weaker answer - Level 1 (3 marks)	
	New technologies help monitor wildlife. Cameras can take pictures of animals at night. GPS collars can track where animals go. Satellites can see deforestation from space. DNA can tell which animals are present. Bat detectors can hear bats. These help conservation because we know where animals are and how many there are. They can be expensive though.	
	What makes the difference	
	• Level 1 lists technologies with no detail of how they work or what data they give; Level 3 links each to the information it provides (territory size, presence, activity).	
	• Level 3's evaluation considers several limitations (calibration, ground-truthing, ethics, species coverage) and reaches a balanced judgement; Level 1 has one word - 'expensive'.	
	• Structure: Level 3 groups by imagery, genetics, sound, position and remote sensing; the list in Level 1 has no organisation.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Success:** Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia – Encarsia parasitic wasps control greenhouse whitefly **up to 2 Problems:** cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators – rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct) – harlequin ladybirds introduced to Europe eat native ladybirds **up to 2 Lesson:** control agents should be specialist feeders on the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest • **endemic** → found naturally only in that one place	4
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Release an artificial sex scent attracting males (or females) of the pest species **1** Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply **1** Agricultural use exceeds medical use, so resistance spreads **1** Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat – hence reduced use of antibiotics is a sustainability goal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
08.4	1 mark for the point below. **1** Grow nectar-rich flowering plants alongside crops – restrict pesticides harmful to bees / spray when not in flower – introduce bee hives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.	
09.1	**1 mark each** 1 mark for each of the 2 points below. **1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people **1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Total impact = population size x the individual (per capita) impact of each person **1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Slash and burn agriculture **1** Fishing **1** Discharge of sewage / biodegradable pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **biodegradable** → broken down by decomposers **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 10% of land is already cultivated and 25% grazed **1** Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies **1** So people with less money lose access to the land and food they used to produce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services **1** So that pesticides and artificial fertilisers become unnecessary rather than just being banned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Unsustainable features:** reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • **Pest control:** cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • **Nutrients:** recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • **Water:** low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • **Soil and carbon:** low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • **Biodiversity:** conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland – agri-environment schemes • **Evaluation:** growing population (perhaps 12 billion) needs more food – lower trophic level diets – obstacles - cost, need for support, gaining acceptance of new techniques – learning from past land degradation (fertile crescent, US mid-west) • **Judgement:** no single strategy sufficient – combination of ecological principles and new technology – justified conclusion **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **salinisation** → salt building up in the soil / soil becoming salty • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Present agriculture relies on inputs that cannot last: chemical pesticides to which pests evolve resistance and which harm non-target species, rock phosphate that is non-renewable, nitrate fertiliser made with fossil-fuel energy, and irrigation that over-draws rivers and groundwater and salinises soil, while ploughing loses soil organic matter and livestock, rice and machinery release greenhouse gases. Each can be replaced by strategies based on ecological principles. Pests are controlled culturally by weeding, mulching, rotation, barrier crops and culling, by biological control and predator habitats, by polyculture and companion crops, and by integrated control that uses chemicals only as a last resort. Nutrients come from recycled organic matter, rotation with legumes and conserved soil biota, so artificial fertiliser, especially nitrate, can be reduced. Water is saved with low water-use crops, reservoirs and aquifer recharge and drip irrigation; soil carbon is kept by low tillage and less machinery, with renewable energy for what remains; methane falls with rice varieties that allow earlier drainage and with ground, carbohydrate-rich cattle feed. Habitats of crop wild relatives, seed banks, hedgerows, ditches, ponds and woodland and agri-environment schemes protect the biodiversity that supplies these services.	
	Food demand will rise with a population that may reach 12 billion, there is little new land, and imports can reduce food availability where it is grown, so yield cannot be ignored. Some strategies cost more or need farmers to accept new techniques, and past land degradation from the fertile crescent to the American mid-west shows what happens when they are not adopted. No single strategy suffices; a combination of ecological principles with new technology is the only route to feeding more people without exhausting the resources food depends on.	
	Weaker answer - Level 1 (2 marks)	
	To make agriculture sustainable farmers should use fewer chemicals. Organic farming does not use pesticides or artificial fertilisers. Crop rotation and manure keep the soil fertile. Farmers should not waste water. Hedges should be kept for wildlife. This would make farming more sustainable but it might produce less food.	
	What makes the difference	
	• Level 1 gives general advice; Level 3 identifies each unsustainable feature and matches it with specific strategies, showing understanding of why they work.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation considers population growth, land limits, cost, adoption and historical evidence; Level 1's is a single clause.	
	• Note that Level 3 avoids defining organic farming as 'not using chemicals' - the textbook stresses it is the positive use of natural processes.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
11.1	**Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why monitor:** species present, population size, density, distribution, survival, age structure, breeding rates, movements – baseline and evaluation – Red List categorisation and quotas rely on it • **Standard techniques and their limitations:** quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR – sampling design (random / systematic, size, number, timing, standardisation) • **New technologies:** satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • **How they improve validity:** representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • **Examples:** penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked	25

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Question	Marking guidance	Marks
	• **Limitations:** cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	
11.2	**Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations – positive feedback and tipping points caused by humans • **Energy:** solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process) – renewable energy, heat pumps, bioleaching, fuel cells • **Materials:** cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals – recycling, Cradle to Cradle, industrial symbiosis • **Diversity and stability:** rainforests / reefs vs agroecosystems – diverse energy mixes • Optimum vs maximum production – supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics – limits and judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	

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